Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Single-cell imaging and data-driven prediction of antibiotic mechanism, resistance, and treatment response			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		Bio-Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Small-Molecule Pharmaceuticals	Bio-processes/ Bio-equipment	Bio-Analytical Instruments
Concept of Proposed Technology	The proposed technology is a single-cell imaging and data-driven analytics platform for mechanistic evaluation of antibiotics and prediction of bacterial treatment response. Conventional antibiotic assays rely on population-averaged measurements and cannot capture cellular heterogeneity, transient stress responses, or rare survivor and persister subpopulations. This platform integrates microfluidic culture of bacterial cells, live single-cell imaging, and quantitative image analysis to monitor changes in nucleoid organization, cell morphology, and intracellular dynamics upon antibiotic exposure. Beyond mechanistic characterization, the resulting single-cell datasets will be used to train predictive models that associate specific imaging-derived phenotypes with antibiotic susceptibility or tolerance. By learning characteristic cellular response signatures, the platform enables data-driven inference of which antibiotics are most likely to be effective against a given bacterial population. Potential applications include early-stage antibiotic screening, resistance-risk assessment, and extension toward			

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	rapid, single-cell–based treatment guidance using patient-derived bacterial samples. The platform is designed as a reusable, SME-friendly technology that supports both antibiotic development and precision antimicrobial strategies.
Rationale for Joint R&D Project	Antibiotic development and clinical deployment both suffer from limited mechanistic resolution and delayed detection of resistance or tolerance. Korean SMEs developing antimicrobial agents or diagnostic technologies may lack access to platforms that can systematically link single-cell bacterial behavior to treatment outcomes. A joint R&D project will combine my group's expertise in single-cell imaging, microfluidics, and quantitative analysis with the SME partner's antibiotic compounds, bacterial isolates, and translational objectives. This collaboration enables generation of high-quality single-cell datasets under controlled antibiotic perturbations and development of predictive models grounded in real biological responses. By transforming imaging data into actionable predictions, the project reduces uncertainty in antibiotic R&D and opens a pathway toward precision treatment strategies. This aligns with innovation, data-driven healthcare, and SME commercialization priorities supported by MSS.
Keywords	1. Antibiotic mechanism of action 2. Single-cell bacterial imaging 3. Antibiotic resistance and persistence 4. Data-driven prediction 5. Microfluidics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objectives of this R&D project are to (1) quantitatively measure nucleoid size, cell morphology, growth dynamics, and intracellular responses over time after antibiotic exposure from phase contrast imaging and fluorescence imaging, (2) identify mechanistic and phenotypic signatures associated with bactericidal action, tolerance, and persistence (ref #1-2), (3) build predictive models trained on single-cell imaging data to classify bacterial responses and infer antibiotic effectiveness, inspired by CellProfiler (ref #3), (4) evaluate feasibility of applying the trained models to bacteria derived from clinically relevant or patient-like samples. Currently, we are capable of live-cell bacterial imaging and quantitatively analyze morphological features from phase-contrast images and nucleoid shape from fluorescence imaging (ref #4). We can also track these features over generations in microfluidics. From imaging data, we can perform statistical and physics-based analysis of heterogeneous populations. For this project, we will develop feature extraction pipelines for bacterial response phenotyping and build predictive models linking single-cell imaging signatures to treatment outcomes.			
R&D Target Specifications	See next section.			
R&D Plan				

Phase	Duration	Key activities	Milestones
1	0-9 months	Imaging bacterial cells under diverse antibiotics	Collect phenotypic signatures depending on the mechanism of action, concentration, and duration of antibiotics.
2	9-15 months	Data analysis and feature extraction	Collect phenotypic signatures depending on the mechanism of action, concentration, and duration of antibiotics.
3	15-21 months	Model training	Predictive response classifier.
4	21-30 months	Validation and testing	Demonstrated treatment prediction and diagnostic power.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	We will design and implement microfluidic and single-cell imaging assays. We will acquire and analyze high-content bacterial response datasets from microscopy. We will define quantitative phenotypic features and response metrics. We will develop and validate predictive models linking imaging data to antibiotic outcomes. We will discuss our results with the Korean SME partner and provide technical guidance.
Expected R&R of Korean SME Partner	We expect the Korean SME partner to participate in experimental design and validation and provide novel antibiotic candidates and bacterial strains or isolates.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project will deliver a validated single-cell imaging and analytics platform that enables Korean SMEs to mechanistically evaluate antibiotics and predict bacterial treatment responses. Expected outcomes include improved antibiotic screening efficiency, early detection of resistance or tolerance, and development of data-driven tools for precision antimicrobial strategies. The platform has strong commercialization potential as an analytical service, screening tool, or foundation for rapid susceptibility assessment technologies, strengthening Korea's antibiotic innovation ecosystem and supporting global competitiveness.

6. References

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(2) Farrar, A., Turner, P., El Sayyed, H., Feehily, C., Chatzimichail, S., Crook, D., ... & Kapanidis, A. (2024). Ribosome Phenotypes Enable Rapid Antibiotic Susceptibility Testing in Escherichia coli. *medRxiv*, 2024-06.

(3) Carpenter, A. E., Jones, T. R., Lamprecht, M. R., Clarke, C., Kang, I. H., Friman, O., ... & Sabatini, D. M. (2006). CellProfiler: image analysis software for identifying and quantifying cell phenotypes. *Genome biology*, *7*(10), R100.

(4) Gray, W. T., Govers, S. K., Xiang, Y., Parry, B. R., Campos, M., Kim, S., & Jacobs-Wagner, C. (2019). Nucleoid size scaling and intracellular organization of translation across bacteria. *Cell*, *177*(6), 1632-1648.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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